

Airline Passenger Analytics using Data Warehouse, ETL, Decision Tree, Bayesian Classification, OLAP Cubes, KNIME, WEKA and Power BI

A CAPSTONE PROJECT REPORT

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DECLARATION

We, **[P. Gayathri, Siva Dakshitha, Akanksha]** of the **[Btech AI\ML]**, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled '**[Airline Passenger Analytics using Data Warehouse, ETL, Decision Tree, Bayesian Classification, OLAP Cubes, KNIME, WEKA and Power BI]**' is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**Airline Passenger Analytics using Data Warehouse, ETL, Decision Tree, Bayesian Classification, OLAP Cubes, KNIME, WEKA and Power BI**” has been carried out by **[P. Gayathri, Siva Dakshitha, Akanksha]** under the supervision of **Dr. D. Beulah David** and is submitted in partial fulfilment of the requirements for the current semester of the B. Tech **[AI\ML]** program at Saveetha Institute of Medical and Technical Sciences, Chennai

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ABSTRACT

Airline passenger data contains valuable information about passenger demographics, travel patterns, booking behavior, flight preferences, and customer satisfaction. However, large volumes of passenger data are difficult to analyze directly because the data may contain missing values, duplicate records, inconsistent formats, and information distributed across different sources. This project develops an airline passenger analytics system using Data Preprocessing, ETL, Data Warehousing, OLAP Cubes, Data Mining, and Visualization techniques.

The project is divided into three major modules. **Module 1 – Data Preprocessing & ETL** focuses on cleaning, transforming, and integrating airline passenger data into a structured format. The ETL process extracts passenger data, performs preprocessing operations, and loads the transformed data into the data warehouse. **Module 2 – Data Warehouse & OLAP** focuses on designing a dimensional data warehouse and creating OLAP cubes for multidimensional analysis of passenger information. **Module 3 – Data Mining & Visualization** applies Decision Tree and Bayesian Classification techniques using KNIME and WEKA to identify passenger patterns and classify passenger-related outcomes. Power BI is used to create interactive dashboards and visual reports.

The proposed system helps transform raw airline passenger data into meaningful information for analysis and decision-making. The combination of ETL, Data Warehouse, OLAP, Data Mining, KNIME, WEKA, and Power BI provides an integrated analytical solution for understanding passenger behavior and improving airline decision-making.

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CHAPTER - 1

INTRODUCTION

1.1 Background Information

The airline industry generates a large amount of passenger-related data through ticket bookings, travel records, passenger profiles, flight information, and customer feedback. Analyzing this data can help airlines understand passenger behavior and identify important patterns.

Traditional methods of handling passenger information may not be sufficient when the volume of data becomes large. Data Warehousing provides a centralized environment where information from different sources can be integrated and stored for analysis.

ETL, which stands for Extract, Transform and Load, is used to collect data from different sources, clean and transform it, and load it into the data warehouse. OLAP technology allows users to analyze warehouse data from multiple dimensions.

Data Mining techniques can further identify hidden patterns and relationships in passenger information. In this project, Decision Tree and Bayesian Classification are used for classification analysis. KNIME and WEKA provide data mining and machine learning environments, while Power BI provides interactive visualization and dashboards.

1.2 Project Objectives

The main objectives of the project are:

1. To collect and preprocess airline passenger data.
2. To perform ETL operations on passenger datasets.
3. To design a structured airline passenger data warehouse.
4. To organize passenger information using dimensional modeling.
5. To create OLAP-based multidimensional analysis.
6. To apply Decision Tree classification to passenger data.
7. To apply Bayesian Classification for passenger prediction and analysis.
8. To use KNIME and WEKA for data mining operations.
9. To develop interactive dashboards using Power BI.
10. To generate meaningful insights from airline passenger data.

1.3 Significance

The project is significant because it provides an integrated approach for analyzing airline passenger information. Instead of examining raw datasets directly, the system organizes data into a structured analytical environment.

The system can help identify passenger trends, compare different passenger categories, analyze travel-related characteristics, and support data-driven decision-making. The combination of data warehousing, OLAP, data mining, and visualization improves the usability of passenger information.

1.4 Scope

The scope of the project includes:

- Airline passenger dataset collection.
- Data cleaning and preprocessing.
- ETL pipeline development.
- Data warehouse design.
- Fact and dimension table creation.
- OLAP cube-based analysis.
- Decision Tree classification.
- Bayesian Classification.
- KNIME-based data analysis.
- WEKA-based classification.
- Power BI dashboard development.
- Analysis of passenger-related patterns.

The project focuses on analytical processing and does not include real-time airline ticket booking or flight-control operations.

1.5 Methodology Overview

The project follows a systematic analytical workflow:

Raw Airline Data → Data Preprocessing → ETL → Data Warehouse → OLAP Cube → Data Mining → Classification → Power BI Visualization → Insights

The first stage prepares the raw data. The second stage stores the processed information in the data warehouse. OLAP provides multidimensional analysis, while Decision Tree and Bayesian Classification identify useful patterns. Finally, Power BI presents the analytical results through interactive dashboards.

CHAPTER - 2

PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

Airline organizations generate large quantities of passenger data. The raw data may contain missing values, duplicate records, inconsistent formats, and different types of attributes.

Analyzing such data manually can be time-consuming and may make it difficult to identify important passenger patterns. Without an integrated analytical system, decisionmakers may not have a clear view of passenger characteristics and trends.

Therefore, there is a need for a structured system that can preprocess passenger data, store it in a data warehouse, perform multidimensional analysis, apply classification algorithms, and present the results through visual dashboards.

2.2 Evidence of the Problem

The major problems associated with raw airline passenger data include:

- Missing or incomplete values.
- Duplicate passenger records.
- Inconsistent data formats.
- Large data volume.
- Difficulty in combining information from multiple sources.
- Limited multidimensional analysis.
- Difficulty in identifying hidden passenger patterns.
- Lack of interactive visualization.

These problems demonstrate the need for a complete data analytics workflow.

2.3 Working Principle

The working principle of the proposed system is based on sequential data processing.

First, airline passenger data is collected from the selected dataset. The data is then cleaned by handling missing values, duplicate records, and inconsistent formats.

The processed data is passed through the ETL process. During extraction, the required data is collected. During transformation, the data is cleaned and converted into a suitable format. During loading, the transformed data is stored in the data warehouse.

The data warehouse is organized into fact and dimension tables. OLAP cubes are then used to perform multidimensional analysis.

The prepared dataset is also provided to Decision Tree and Bayesian Classification algorithms using KNIME and WEKA. Finally, the analytical results are visualized using Power BI dashboards.

2.4 Supporting Data/Research

The project uses concepts from:

- Data Preprocessing
- ETL
- Data Warehousing
- Dimensional Modelling
- OLAP
- Data Mining
- Decision Tree Classification
- Bayesian Classification
- KNIME
- WEKA
- Power BI

These technologies are integrated to create an end-to-end airline passenger analytics system.

CHAPTER - 3

SOLUTION DESIGN AND IMPLEMENTATION

3.1 Development and Design Process

The project development process consists of the following stages:

1. **Data Collection** – Obtain the airline passenger dataset.
2. **Data Preprocessing** – Clean and prepare the raw data.
3. **ETL Processing** – Extract, transform, and load the data.
4. **Data Warehouse Design** – Create a structured warehouse.
5. **OLAP Analysis** – Perform multidimensional analysis.
6. **Data Mining** – Apply classification techniques.
7. **Model Evaluation** – Evaluate Decision Tree and Bayesian models.
8. **Visualization** – Create Power BI dashboards.
9. **Insight Generation** – Interpret the analytical results.

3.2 Tools and Technologies Used

Tool/Technology	Purpose
ETL	Extract, transform and load passenger data
Data Warehouse	Centralized analytical data storage
OLAP Cubes	Multidimensional data analysis
KNIME	Data preprocessing and data mining
WEKA	Classification and machine learning
Decision Tree	Classification and pattern identification
Bayesian Classification	Probabilistic classification
Power BI	Interactive data visualization
Dataset	Source of airline passenger information

3.3 Solution Overview

The proposed solution integrates different data analytics technologies into a single workflow.

Input Layer:

Airline passenger dataset.



Preprocessing Layer:

Missing value handling, duplicate removal, data transformation, and data cleaning.



ETL Layer:

Extract → Transform → Load.



Data Warehouse:

Fact table + Dimension tables.



OLAP Layer:

Slice, Dice, Roll-up and Drill-down operations.



Data Mining Layer:

Decision Tree + Bayesian Classification.



Tools:

KNIME + WEKA.

**Visualization Layer:**

Power BI dashboards.

**Output:**

Passenger insights and analytical reports.

3.4 Engineering Standards Applied

The project follows structured data processing and database design principles. Data is organized systematically to reduce redundancy and improve analytical performance.

The ETL process follows a defined sequence of extraction, transformation, and loading. Dimensional modelling is used to organize analytical data into fact and dimension structures.

Data mining models are developed using established classification techniques, and visualization follows structured dashboard design principles to make analytical information easier to understand.

3.5 Solution Justification

The proposed solution is suitable because each technology addresses a specific requirement of the project.

ETL prepares the raw data, while the data warehouse provides centralized storage. OLAP supports multidimensional analysis. Decision Tree and Bayesian Classification provide predictive and classification capabilities. KNIME and WEKA support data mining implementation, and Power BI provides interactive visualization.

Therefore, the integration of these technologies creates a complete analytical pipeline rather than depending on a single technique.

MODULE 1 – DATA PREPROCESSING & E

Overview

Module 1 focuses on preparing airline passenger data for further analysis. Raw datasets often contain missing values, duplicate records, incorrect formats, and unnecessary attributes.

The objective of this module is to convert raw passenger data into a clean and structured dataset.

Data Collection

The airline passenger dataset is collected and examined to identify the available attributes. The dataset may contain passenger-related information such as:

- Passenger ID
- Gender
- Age
- Customer Type
- Type of Travel
- Class
- Flight Distance
- In-flight Services
- Food and Drink
- Seat Comfort
- Departure/Arrival Information
- Departure Delay
- Arrival Delay
- Customer Satisfaction

The exact attributes depend on the dataset selected for implementation.

Data Preprocessing

The following preprocessing operations are performed:

- Identifying missing values.
- Removing duplicate records.
- Correcting inconsistent values.
- Converting data types.
- Removing unnecessary attributes.
- Encoding categorical attributes where required.
- Checking outliers and abnormal values.
- Preparing the final analytical dataset.

ETL Process

The ETL process consists of three major stages.

Extract

The required airline passenger data is extracted from the source dataset.

Transform

The extracted data is cleaned and transformed. Missing values are handled, duplicate records are removed, categorical values are standardized, and attributes are converted into appropriate formats.

Load

The transformed data is loaded into the data warehouse for further OLAP and data mining analysis.

Output

The output of Module 1 is a **clean, integrated, and structured airline passenger dataset** that can be used by the data warehouse and classification models.

MODULE 2 – DATA WAREHOUSE & OLAP

Overview

Module 2 focuses on storing the processed airline passenger information in a data warehouse and performing multidimensional analysis using OLAP concepts.

Data Warehouse Design

The data warehouse provides a centralized repository for the processed passenger data.

A possible dimensional structure includes:

Fact Table – Passenger Analytics Fact

The fact table can contain measurable passenger-related values such as:

- Passenger count

- Flight distance
- Departure delay
- Arrival delay
- Service ratings
- Satisfaction measure

Dimension Tables

Possible dimensions include:

- **Passenger Dimension**
- **Travel Dimension**
- **Flight Dimension**
- **Date Dimension**
- **Service Dimension**

The exact structure can be adjusted according to the selected dataset.

Dimensional Modelling

A star-schema approach can be used for organizing the data warehouse.

The central fact table contains measurable values, while dimension tables provide descriptive information for analysis.

Dimension Tables → Fact Table → OLAP Analysis

This structure supports efficient analytical queries.

OLAP Cube

An OLAP cube allows passenger information to be analyzed across multiple dimensions.

For example:

Passenger Satisfaction × Travel Type × Class

or

Passenger Count × Customer Type × Class

OLAP operations include:

Roll-Up

Roll-up summarizes detailed information into a higher-level category.

Drill-Down

Drill-down moves from summarized information to detailed information.

Slice

Slice selects a specific value from one dimension.

Dice

Dice selects a smaller sub-cube using multiple dimension conditions.

Output

The output of Module 2 is a structured **airline passenger data warehouse and OLAP analytical environment** that allows multidimensional analysis.

MODULE 3 – DATA MINING & VISUALIZATION

Overview

Module 3 focuses on extracting useful patterns from airline passenger data using data mining algorithms and presenting the results through visualization.

The major techniques used are:

- Decision Tree Classification
- Bayesian Classification
- KNIME
- WEKA
- Power BI

Decision Tree Classification

A Decision Tree is a classification technique that represents decisions in the form of a tree structure.

The tree consists of:

- Root node
- Decision nodes
- Branches
- Leaf nodes

The algorithm selects important attributes for splitting the data and eventually assigns a class to each record.

For airline passenger analytics, the model can be used to classify passenger-related outcomes such as satisfaction categories, depending on the selected target attribute.

Bayesian Classification

Bayesian Classification is based on probability and Bayes' theorem.

The general form is:

$$P(C|X) = [P(X|C) \times P(C)] / P(X)$$

Where:

- C = class
- X = observed data
- $P(C|X)$ = probability of class C given X
- $P(X|C)$ = probability of X given class C
- $P(C)$ = prior probability of class C
- $P(X)$ = probability of observed data

The classifier uses these probabilities to assign passenger records to the most appropriate class.

KNIME

KNIME is used to create a visual data analytics workflow.

A typical workflow can contain:

File Reader → Data Preprocessing → Partitioning → Decision Tree/Bayesian Model → Predictor → Scorer

The workflow allows data preparation, model development, prediction, and evaluation.

WEKA

WEKA can be used to perform machine learning and classification experiments.

The passenger dataset can be imported into WEKA, preprocessed, and supplied to classification algorithms. The resulting model can be evaluated using measures such as:

- Accuracy
- Precision
- Recall
- F-Measure
- Confusion Matrix

Power BI Visualization

Power BI is used to convert analytical results into interactive dashboards.

Possible visualizations include:

- Passenger distribution by class.
- Passenger distribution by travel type.
- Customer type analysis.
- Satisfaction analysis.
- Flight distance analysis.
- Delay analysis.
- Service-rating analysis.
- Classification results.

Interactive filters and slicers can be provided to allow users to explore passenger information.

Output

The output of Module 3 is a combination of **classification models, model evaluation results, and interactive Power BI dashboards**.

CHAPTER - 4

RESULTS AND RECOMMENDATIONS

4.1 Evaluation of Results

The developed system provides an integrated method for analyzing airline passenger information.

The preprocessing stage produces a cleaner dataset. ETL integrates the prepared data into the warehouse. OLAP enables multidimensional analysis, while Decision Tree and Bayesian Classification provide classification capabilities.

KNIME and WEKA support the implementation and evaluation of data mining models. Power BI converts the analytical results into interactive dashboards.

The results should be evaluated using the actual experimental values obtained from the implemented dataset. Important evaluation measures include classification accuracy, precision, recall, F-measure, and confusion matrix.

4.2 Challenges Encountered

The major challenges include:

1. Handling missing values.
2. Removing duplicate records.
3. Managing categorical attributes.
4. Selecting relevant attributes.
5. Designing an appropriate warehouse structure.
6. Connecting different analytical tools.
7. Selecting suitable classification algorithms.
8. Interpreting classification results.
9. Designing meaningful dashboards.

4.3 Possible Improvements

Future improvements can include:

- Using larger airline datasets.
- Adding real-time passenger data.
- Applying additional machine learning algorithms.
- Improving prediction accuracy.
- Integrating automated ETL pipelines.
- Adding real-time Power BI reporting.
- Using advanced predictive analytics.

- Incorporating additional passenger and flight dimensions.

4.4 Recommendations

The system can be recommended for analytical applications where airline passenger data needs to be integrated and analyzed systematically.

Airline organizations can use the analytical approach to understand passenger characteristics, compare travel categories, analyze satisfaction patterns, and support data-driven planning.

CHAPTER - 5

REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

5.1 Key Learning Outcomes

Through this project, knowledge was gained in:

- Data preprocessing.
- ETL operations.
- Data warehouse development.
- OLAP analysis.
- Data mining.
- Classification algorithms.
- KNIME.
- WEKA.
- Power BI visualization.

5.2 Academic Knowledge

The project helped connect theoretical concepts from data warehousing and data mining with practical implementation.

Concepts such as ETL, dimensional modelling, OLAP, classification, and data visualization were applied to a real-world airline analytics problem.

5.3 Technical Skills

The project improved technical skills in:

- Dataset preparation.
- Data transformation.
- Data warehouse design.
- OLAP analysis.
- Decision Tree implementation.
- Bayesian Classification.
- KNIME workflow development.
- WEKA analysis.
- Power BI dashboard creation.

5.4 Problem-Solving and Critical Thinking

The project required identifying data quality problems and determining appropriate preprocessing methods. It also required selecting suitable analytical techniques and interpreting the outputs of different classification models.

Comparing different analytical approaches improved problem-solving and critical-thinking abilities.

5.5 Conclusion of Personal Development

The project provided practical experience in handling a complete data analytics workflow. It improved understanding of how raw data can be transformed into useful information through preprocessing, warehousing, mining, and visualization.

CHAPTER - 6

CONCLUSION

The **Airline Passenger Analytics using Data Warehouse, ETL, Decision Tree, Bayesian Classification, OLAP Cubes, KNIME, WEKA and Power BI** project provides an integrated framework for analyzing airline passenger information.

The project combines three major modules: **Data Preprocessing & ETL, Data Warehouse & OLAP, and Data Mining & Visualization.**

Module 1 prepares and transforms the raw passenger data. Module 2 stores the processed data in a structured data warehouse and enables multidimensional OLAP analysis. Module 3 applies Decision Tree and Bayesian Classification using KNIME and WEKA and presents the results using Power BI.

Overall, the proposed system demonstrates how data warehousing and data mining technologies can be combined to transform raw airline passenger data into meaningful analytical information.

6.1 Future Enhancement

The project can be enhanced by integrating real-time airline data sources, advanced machine learning algorithms, automated ETL processes, real-time dashboards, and predictive passenger analytics. Future versions can also incorporate additional data sources such as booking history, loyalty-program information, and real-time flight information.

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APPENDICES

APPENDIX A – AIRLINE PASSENGER DATASET

The airline passenger dataset is used as the primary source of data for the proposed analytics system. The dataset contains passenger-related information such as passenger demographics, customer type, type of travel, travel class, flight distance, service ratings, departure delay, arrival delay, and passenger satisfaction. The dataset provides the required information for data preprocessing, ETL, data warehousing, data mining, classification, and visualization.

APPENDIX B – DATA PREPROCESSING

The raw airline passenger dataset is preprocessed before performing further analysis. The preprocessing stage includes identifying and handling missing values, removing duplicate records, correcting inconsistent data, converting attributes into suitable data types, and selecting relevant attributes. The cleaned dataset provides consistent and reliable information for the subsequent ETL and data mining processes.

APPENDIX C – ETL PROCESS

The ETL process is used to transfer airline passenger data from the source dataset into the data warehouse. The process consists of three stages: Extract, Transform, and Load. During extraction, the required passenger data is collected from the source. During transformation, the data is cleaned, standardized, and converted into an appropriate format. During loading, the transformed data is stored in the data warehouse for analytical processing.

APPENDIX D – DATA WAREHOUSE DESIGN

The data warehouse provides a centralized repository for storing the processed airline passenger information. A dimensional data warehouse structure is used to organize data for analytical operations. The warehouse contains a central passenger analytics fact table connected with appropriate dimension tables such as Passenger, Travel, Flight, Date, and Service dimensions.

APPENDIX E – FACT AND DIMENSION TABLES

The fact table stores measurable airline passenger information, including passenger count, flight distance, departure delay, arrival delay, service ratings, and satisfaction measures. The dimension tables provide descriptive information about passengers, travel characteristics, flights, dates, and services. The combination of fact and dimension tables supports efficient analytical queries and multidimensional analysis.

APPENDIX F – OLAP CUBE ANALYSIS

The OLAP cube is used to analyze airline passenger information across multiple dimensions. OLAP operations such as roll-up, drill-down, slice, and dice are applied to obtain different levels of analytical information. These operations enable passenger data to be summarized, detailed, filtered, and compared according to different dimensions such as travel type, class, customer type, and satisfaction.

APPENDIX G – KNIME WORKFLOW

KNIME Analytics Platform is used to develop data analytics workflow for the airline passenger dataset. The workflow includes data reading, preprocessing, data partitioning, classification model development, prediction, and evaluation. Decision Tree and Bayesian Classification techniques are implemented through the KNIME workflow to analyze passenger-related patterns.

APPENDIX H – DECISION TREE CLASSIFICATION

The Decision Tree classification technique is applied to identify passenger-related patterns and classify passenger records according to the selected target attribute. The Decision Tree consists of a root node, decision nodes, branches, and leaf nodes. The classification model uses relevant passenger attributes to make decisions and generate corresponding class prediction.

APPENDIX I – BAYESIAN CLASSIFICATION

Bayesian Classification is applied to classify airline passenger records using probability-based analysis. The method uses Bayes' theorem to calculate the probability of a particular class based on the observed passenger attributes. The classification results are evaluated to determine the effectiveness of the model in predicting the selected passenger category.

The general form of Bayes' theorem is:

$$P(C|X) = [P(X|C) \times P(C)] / P(X)$$

Where C represents the class and X represents the observed passenger attributes.

APPENDIX J – WEKA IMPLEMENTATION

WEKA is used for implementing and evaluating the data mining algorithms. The airline passenger dataset is imported into WEKA and preprocessed before classification. Decision Tree and Bayesian Classification algorithms are applied to the prepared dataset. The classification results are evaluated using accuracy, precision, recall, F-measure, and the confusion matrix.

APPENDIX K – CLASSIFICATION RESULTS

The classification results obtained from Decision Tree and Bayesian Classification are analyzed and compared. The evaluation considers classification accuracy and other performance measures such as precision, recall, F-measure, and confusion matrix. The comparison helps identify the effectiveness of the selected classification techniques for airline passenger analytics.

APPENDIX L – POWER BI DASHBOARD

Power BI is used to develop interactive dashboards for airline passenger analytics. The dashboard presents passenger information through charts, graphs, tables, cards, and interactive filters. The visualization includes passenger distribution by class, customer type, travel type, satisfaction, flight distance, delays, and service-related attributes.

APPENDIX M – DATA VISUALIZATION

The data visualization component represents the processed airline passenger information in an understandable graphical form. Various charts and visual elements are used to identify passenger distributions, travel patterns, satisfaction levels, flight characteristics, delay patterns, and service-related trends. The visualizations provide a clear representation of the analytical results.

APPENDIX N – COMPLETE PROJECT WORKFLOW

The complete airline passenger analytics workflow integrates all three project modules. The process begins with the airline passenger dataset and continues through data preprocessing, ETL, data warehouse development, OLAP analysis, data mining, classification, and visualization.

Airline Passenger Dataset → Data Preprocessing → ETL → Data Warehouse → OLAP Cube → Decision Tree & Bayesian Classification → KNIME & WEKA → Power BI Visualization → Passenger Analytics

APPENDIX O – FINAL PROJECT OUTPUT

The final output of the project is an integrated airline passenger analytics system. The system produces a cleaned passenger dataset, processed ETL data, a structured data warehouse, OLAP-based analytical results, Decision Tree and Bayesian Classification models, KNIME and WEKA outputs, and interactive Power BI dashboards. These outputs provide meaningful insights into airline passenger characteristics, travel patterns, satisfaction, and related analytical measures.